Work in progress — draft, not peer reviewed

# A Lean 4 Formalization of the Classical Continuation Chain for Periodic Navier–Stokes on $\mathbb{T}^3$: Energy Identity, $H^3$ Commutator Reduction and the BKM Route

*Companion paper to* `BenFrohman/periodic-ns-classical-continuation`

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**How to cite / What is claimed.** In Lean 4 with Mathlib, we give a machine-checked reduction of Clay Millennium Statement (B) (periodic Navier–Stokes on $\mathbb{T}^3$) to the single hypothesis `PeriodicBKMIntegrability`. We prove this hypothesis is equivalent to global existence of smooth solutions for every smooth periodic initial field. The BKM criterion and $H^3$ control are proved unconditionally, using only the standard axioms. `PeriodicBKMIntegrability` remains open. The equivalence of vorticity integrability with continuation of smooth solutions is classical (Beale–Kato–Majda, 1984 [1]); the novelty here is the complete, `sorry`-free formalization. $\mathbb{T}^3$ is modelled as $\mathbb{R}^3$ with $\mathbb{Z}^3$-periodicity and unit-cell integrals (Section 2). Axiom audit at commit `36fa434`: 1048 theorems, 285 definitions, only `propext`, `Classical.choice`, `Quot.sound` (Appendix A). Cite as: B. S. Frohman, reduction paper, DOI 10.5281/zenodo.23842295; main paper concept DOI 10.5281/zenodo.23842555, this version (v11) DOI 10.5281/zenodo.24091995.

**Draft — tracks build in progress.** This document reports exactly what the Lean development proves at the stated commit. It does *not* solve the Clay Millennium problem, and does not prove global regularity of the 3D Navier–Stokes equations. The step `PeriodicBKMIntegrability` is the open Clay core and remains an explicit hypothesis.

**Abstract**

We describe a Lean 4/Mathlib [4, 15] formalization of the classical continuation steps for smooth divergence-free solutions of the incompressible Navier–Stokes equations on the unit torus $\mathbb{T}^3$, along the Statement (B) route [5] of the companion repository `fefferman-clay-navier-stokes`. Proved with no hypothesis Props: local existence for $\nu \geq 0$, uniqueness via an $L^2$ energy-difference estimate, the Fourier coefficient ODE, coefficient decay, the $H^s$ energy identity, Parseval's identity on the unit cube, and the Kato–Ponce toolkit [7] at $s = 3$ (integration by parts, Leibniz rule, a Gagliardo–Nirenberg inequality, a bound on every lower-order term, and vanishing of the top-order term from $\operatorname{div} u = 0$). Also proved: the $s = 3$ physical-space commutator bound `periodicPhysicalCommutatorBound`. Also proved: the low-frequency vorticity bound `periodicLowFreqVorticityBound`, and hence, with no hypothesis Props, $H^3$ Beale–Kato–Majda control (`periodicBKMHsControl_three`) and the periodic BKM criterion (`periodicBKMCriterion`). Statement (B) is proven conditionally on `PeriodicBKMIntegrability`, which is open (`clay_B_exact_of_integrability`). Small zero-mean $H^1$ data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_bending` gives integrability from `PeriodicDirectionEnstrophyGrowthL1`, `PeriodicH1BlowupCriterion` and the open `PeriodicBendingRateHiddenProperty`.


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*Keywords.* Navier–Stokes equations; periodic domain; Beale–Kato–Majda criterion; continuation; Lean 4; Mathlib; formal verification; work in progress.

# 1 Introduction

The incompressible Navier–Stokes equations on the three-dimensional torus $\mathbb{T}^3$ are the setting of Statement (B) of the Clay Millennium Problem [5]. Their classical continuation theory—local existence, uniqueness, the $H^s$ energy identities and the Beale–Kato–Majda (BKM) criterion [1]—is well established on paper. The purpose of this work is to formalize that theory in Lean 4 with Mathlib [4, 15], and to record precisely which steps are machine-checked without hypotheses, which are machine-checked relative to explicitly named hypotheses, and which remain open.

Section 5 collects the results proved with no hypothesis Props. Section 6 records the conditional results, culminating in the reduction of Statement (B) to the single open hypothesis `PeriodicBKMIntegrability`. Section 7 documents the further reductions of that hypothesis obtained so far, and Section 8 summarizes the status of every named Prop. No result in this paper resolves the Clay problem.

# 2 Modelling of the torus

In the Lean development the torus is not a quotient type. The carrier `T3` is `Euclidean Space ℝ (Fin 3)`, i.e. $\mathbb{R}^3$; a field on $\mathbb{T}^3$ is a field on $\mathbb{R}^3$ satisfying `IsZ3Periodic` (invariance under translation by $\mathbb{Z}^3$), and every spatial integral is taken over one unit cell $[0,1]^3$ (`torus Integral`). `IsPeriodicSmoothSolution` (`FeffermanStatements.lean`) requires a smooth $\mathbb{Z}^3$-periodic pair $(u, p)$ with

$$\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p, \qquad \operatorname{div} u = 0, \qquad u(0) = u_0.$$

This is a faithful encoding of the periodic problem; every Clay- and BKM-type statement in this paper is to be read in this model.

# 3 Conventions and naming

Every hypothesis is a named `Prop` passed as an explicit theorem argument; no `axiom`, `sorry` or `admit` appears. The axiom audit (`lake env lean AxiomAudit.lean`) reports only `propext`, `Classical.choice` and `Quot.sound`. Lean identifiers defined in this repository are *coined in this build* (marked †), not names from the author's May/June 2026 manuscript. The identifiers `Clay_B_exact`, `PeriodicLocalExistence`, `PeriodicBKMCriterion` and `PeriodicBKM Integrability` come from the dependency `fefferman-clay-navier-stokes` (pinned at `ebcf109`).

# 4 Relation to the May/June 2026 manuscript

The author's manuscript frames regularity through a tethered Lyapunov functional $S_\varepsilon$ and a comparison ODE

$$y' = Cy^2 - \kappa'' y^3, \tag{4.1}$$

with transport cancellation, viscous dissipation, a Hölder–Sobolev gate and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). For *unmodified* Navier–Stokes ($\kappa = 0$) the cubic damping is

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absent, and the remaining gap is hypothesis (H5): a non-circular majorant for the vorticity. In this repository that gap appears exactly as `PeriodicBKMIntegrability`, i.e.

$$\int_0^T \|\omega(t)\|_{L^\infty}\,\mathrm{d}t < \infty. \tag{4.2}$$

Nothing here closes it. The transport cancellation of the manuscript corresponds to `top_term_zero` below.

## 5 Proved results (no hypothesis Props)

**Theorem 5.1** (Local existence). `periodicLocalExistence_of_nonneg`$^\dagger$*: for $\nu \geq 0$,* `PeriodicLocalExistence` $\nu$ *holds (Galerkin method at $s = 3$).*

**Theorem 5.2** (Energy difference / uniqueness). `periodicEnergyDifferenceEstimate`$^\dagger$*: for $\nu \geq 0$ and two classical solutions $u, v$ with equal data,*

$$E(t) = \int_{[0,1]^3} |u - v|^2\,\mathrm{d}x \quad \textit{satisfies} \quad E'(t) \leq 2C\,E(t).$$

*With Grönwall (*`eq_zero_of_deriv_le_mul`$^\dagger$*,* `periodicVelocityUniqueness_of_energy`$^\dagger$*) this gives uniqueness.*

**Theorem 5.3** (Fourier coefficient ODE, decay, $H^s$ energy identity). `periodicFourierCoeffODE`$^\dagger$, `periodicFourierCoeffDecay`$^\dagger$ *and* `periodicFourierHsEnergyIdentity`$^\dagger$ *(file* `CoeffODE.lean`*) prove the Props* `PeriodicFourierCoeffODE`, `PeriodicFourierCoeffDecay`, `PeriodicFourierHsEnergyIdentity` *for every $\nu$ (and every $s$): each Fourier coefficient $\hat{u}_k(t)$ obeys the projected ODE, coefficients decay faster than any power, and*

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{H^s}^2 = -2\nu \sum_k (1 + |k|^2)^s\, 4\pi^2 |k|^2 |\hat{u}_k|^2 + \big(\textit{nonlinear rate}\big). \tag{5.1}$$

*The bridge is* `periodicFourierHsEnergyIdentity_of_coeff`$^\dagger$ *(*`HsEnergyIdentity.lean`*).*

**Theorem 5.4** (Parseval on the unit cube). `hasSum_prod_cubeCoeff`$^\dagger$ *(*`Parseval.lean`*): for continuous periodic $f, g$,*

$$\sum_{k \in \mathbb{Z}^3} \overline{\hat{f}_k}\,\hat{g}_k = \int_{[0,1]^3} \bar{f}\,g\,\mathrm{d}x,$$

*via* `torusLift`$^\dagger$ *and Mathlib's torus Parseval.*

**Theorem 5.5** (Physical form of the $H^3$ rate). `hasSum_rate_three`$^\dagger$*: the Fourier-side $H^3$ nonlinear rate is summable and equals*

$$-2\sum_i \int \Big(1 - \frac{\Delta}{4\pi^2}\Big)^3 v_i \left((v \cdot \nabla)v\right)_i \mathrm{d}x.$$

**Theorem 5.6** (Kato–Ponce toolkit, $s = 3$; `KatoPonce.lean`). *Proved: unit-cell integration by parts* `cellI_ibp`$^\dagger$, `cellI_ibp'`$^\dagger$*; Leibniz rule* `DL_mul`$^\dagger$*; Gagliardo–Nirenberg on the cell* `gagliardoNirenberg_cell`$^\dagger$ *and* `cellI_sq_mul_abs_le`$^\dagger$*; and*

- `lower_term_bound`$^\dagger$*: every lower-order Leibniz term (first factor of order $\geq 1$) satisfies*

$$\left|\int D^L v_i \cdot D^{p_1} v_l\, D^{p_2} \partial_l v_i\,\mathrm{d}x\right| \leq 5\|\nabla v\|_{L^\infty}\,Q(v),$$

  *with $Q =$* `physQ`$^\dagger$*;*


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- `top_term_zero`$^\dagger$: $\sum_{i,l}\int D^L v_i\, v_l\, D^L\partial_l v_i\,\mathrm{d}x = 0$, *because* $\operatorname{div} v = 0$ *(transport cancellation).*

**Theorem 5.7** (Continuation alternative)**.** `periodicBKMCriterion_of_hsControl`$^\dagger$*: for* $\nu \geq 0$*,* $s > 5/2$*,* `PeriodicBKMHsControl` $\nu$ $s$ *implies* `PeriodicBKMCriterion` $\nu$ *(the* $H^s$ *blow-up alternative is proved).*

## 6 Conditional results

**Named hypothesis 6.1** (`PeriodicPhysicalCommutatorBound` $C^\dagger$, $s = 3$)**.** For smooth periodic divergence-free $v$,

$$-2\sum_i \int \Big(1 - \frac{\Delta}{4\pi^2}\Big)^3 v_i \left((v\cdot\nabla)v\right)_i \mathrm{d}x \leq C\|\nabla v\|_{L^\infty}\|v\|_{H^3}^2. \tag{6.1}$$

*Status:* **proved** in `CommutatorAssembly.lean`: `periodicPhysicalCommutatorBound`$^\dagger$ gives it with $C = 2\cdot 64\cdot 315\cdot 64\,(4\pi^2)^3$, from `lower_term_bound`, `top_term_zero`, `word_bound`$^\dagger$ and `physQ_le_hsSq`$^\dagger$. No longer a hypothesis.

**Named hypothesis 6.2** (`PeriodicLowFreqVorticityBound`$^\dagger$)**.** There is $K \geq 0$ such that for all $R \geq 2$,

$$\|V_R \nabla v\|_{L^\infty} \leq K(1 + \log R)\|\omega\|_{L^\infty}. \tag{6.2}$$

*Status:* **proved** in `LowFreqAssembly.lean` (`periodicLowFreqVorticityBound`$^\dagger$). The proof writes the Biot–Savart law in Riesz form,

$$\widehat{\partial_j v_i} = q_{j,i+2}\,\hat{\omega}_{i+1} - q_{j,i+1}\,\hat{\omega}_{i+2}, \qquad q_{jb} = \frac{k_j k_b}{|k|^2}.$$

It splits $q$ into a low piece and $S = \lceil \log_2 R\rceil + 3$ smooth dyadic blocks, each with a uniformly $L^1$-bounded kernel (`DyadicBlock.lean`). Every piece is followed by the uniformly $L^1$-bounded de la Vallée Poussin kernel (`VPKernel.lean`). No longer a hypothesis.

**Named hypothesis 6.3** (`PeriodicBKMIntegrability` $\nu$)**.** From the dependency `fefferman-clay-navier-stokes`: $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$ on every finite interval of existence. *Status:* **open**. This is the Clay core and the manuscript's (H5) gap.

**Proposition 6.4.** `periodicFourierCommutatorBound_of_physical`$^\dagger$*:* `PeriodicPhysicalCommutatorBound` $C \Rightarrow$ `PeriodicFourierCommutatorBound` $3$ $C$*.* `periodicBKMHsControl_of_physical_lowFreq`$^\dagger$*:* $(\exists C \geq 0,$ `PeriodicPhysicalCommutatorBound` $C)$ *and* `PeriodicLowFreqVorticityBound` *give* `PeriodicBKMHsControl` $\nu$ *3. Combined with the proved bound:* `periodicBKMHsControl_three_of_lowFreq`$^\dagger$ *gives* `PeriodicBKMHsControl` $\nu$ *3 from* `PeriodicLowFreqVorticityBound` *alone.*

**Theorem 6.5** (Unconditional corollaries, `BKMFinal.lean`)**.** `periodicBKMHsControl_three`$^\dagger$*:* `PeriodicBKMHsControl` $\nu$ *3 for every* $\nu$*, with no hypotheses.* `periodicBKMCriterion`$^\dagger$*:* `PeriodicBKMCriterion` $\nu$ *for every* $\nu$*, with no hypotheses.*

**Proposition 6.6** (Statement (B), conditional)**.** `clay_B_exact_of_hsControl`$^\dagger$*: for* $s > 5/2$*,* `PeriodicBKMHsControl` $\nu$ $s$ *and* `PeriodicBKMIntegrability` $\nu$ *give* `Clay_B_exact` $\nu$*. Also* `clay_B_exact_of_corrected_route'`$^\dagger$ *and* `clay_B_exact_of_bkm`$^\dagger$*. The final form,* `clay_B_exact_of_integrability`$^\dagger$*, is conditional* only *on the open* `PeriodicBKMIntegrability` $\nu$*.*

*Remark* 6.7 (A failed route). A Wiener-algebra/Fourier commutator estimate was tried and abandoned: its low-frequency part grows like $R^{3/2}$, not $\log R$. The $s = 3$ physical-space estimate replaces it.


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# 7 Further reductions of the open hypothesis (work in progress)

Files `IntegrabilityAttack.lean`, `SmallDataAttack.lean`, `TetherClaimAttack.lean`, `EnstrophyIdentity.lean`, `TetherKappaZero.lean`, `DirectionCriterion.lean` (commits `f981e71`, `052cef1`, `ea5133d`, `0cd3551`, `a35d13a`); the file names are historical. All identifiers in this section are coined. **This does not prove `PeriodicBKMIntegrability`.**

- `integrableOn_vorticity_of_lt_tstar` (proved): if $T < T^*$, then $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$.
- `periodicBKMIntegrability_iff_globalLifespan` (proved): `PeriodicBKMIntegrability` $\nu$ holds iff `PeriodicGlobalLifespan` $\nu$ ($T^* = \infty$ for all smooth data). The open hypothesis is therefore exactly global existence.
- Route 2 (proved from named classical hypotheses, not formalized here):
  - `integrability_of_smallDataGlobal`, from `PeriodicSmallDataGlobal` [6];
  - `integrability_of_prodiSerrin`, from `PeriodicProdiSerrinCriterion` [10, 12] (endpoint $L_t^\infty L_x^\infty$) plus a velocity bound.
- Route 1 (the $\kappa = 0$ analogue of the tether), `periodicBKMIntegrability_of_vorticityAPriori` (proved): from the **open** `PeriodicVorticityAPrioriBound`. It breaks because with no damping term the enstrophy satisfies only $E' \le CE^3$ and the max-point bound gives $Y' \lesssim Y^2 \log(\cdot)$. Neither gives a uniform bound.
- `periodicEnstrophyIdentity` (`EnstrophyIdentity.lean`, proved, **no hypotheses**): the $\mathbb{T}^3$ enstrophy ($H^1$) energy identity, the $s = 1$ case of the $H^s$ identity (5.1).
- Small zero-mean $H^1$ data (`EnstrophyIdentity.lean`): `riccati_barrier_Ioo` (proved) and `periodicTstar_eq_top_of_smallH1_of_rate`, proved from the unconditional identity plus the named classical hypotheses `PeriodicEnstrophyRateBound` (Ladyzhenskaya/Poincaré trilinear bound [8]) and `PeriodicH1BlowupCriterion` (Leray [9]), neither of which was formalized at that stage. Subsequently, the trilinear bound `PeriodicH1TrilinearBound` is now proved (`periodicH1TrilinearBound`, `Ladyzhenskaya.lean`, explicit unoptimized constant, Fourier/Agmon route to a Ladyzhenskaya-type estimate; the inequality is classical, O. A. Ladyzhenskaya 1958 in 2D and her 1961 monograph in 3D [8]), as is `PeriodicMeanConservation` (`periodicMeanConservation`); hence `periodicTstar_eq_top_of_smallH1_ladyzhenskaya` depends only on the named `PeriodicH1BlowupCriterion`, which remains **not formalized**. This does not resolve the Clay problem. *Correction.* The earlier hypothesis `PeriodicEnstrophyRiccati` with $\alpha > 0$ (`SmallDataAttack.lean`, used by `periodicTstar_eq_top_of_smallH1`, `integrability_of_smallH1`) is false for small constant flows, since the mean mode is conserved; the zero-mean rate-bound version replaces it.
- Tether claim (`TetherClaimAttack.lean`): `cubic_riccati_barrier` (proved); `periodicVorticityAPrioriBound_of_tetherCubicRate`, `clay_B_exact_of_tetherCubicRate` proved from the **open** `PeriodicTetherCubicRate` (not derived from unmodified NS); `uniform_cubic_rate_false_of_scaling` (proved, abstract scaling step: a data-uniform cubic rate is contradictory).
- Direction route: `integrability_of_directionCriterion`, `periodicBKMIntegrability_of_direction`, proved from the classical `PeriodicDirectionCriterion` (Beirão da Veiga–Berselli [2] / Constantin–Fefferman [3], not formalized) plus the **open** `PeriodicDirectionRegularity`. In addition, `periodicDirectionCriterion_of_growth` proves `PeriodicDirectionCriterion` from the unconditional enstrophy identity, the linear


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Grönwall lemma `exp_gronwall_Ioo`, and two named open hypotheses: `PeriodicDirectionEnstrophyGrowth` (coined; Constantin–Fefferman 1993 geometric depletion of the strain kernel plus Beirão da Veiga–Berselli 2002, constant-rate form, not formalized) and `PeriodicH1BlowupCriterion` (Leray 1934). `periodicBKMIntegrability_of_directionGrowth` still needs the open `PeriodicDirectionRegularity`.

- Bending rate (`BendingRate.lean`): `hasDerivAt_vortDir_lagrangian` (proved) gives the pointwise identity

$$\frac{D\mathbf{t}}{Dt} = (I - \mathbf{t}\otimes\mathbf{t})S\mathbf{t} + \nu|\omega|^{-1}(I - \mathbf{t}\otimes\mathbf{t})\Delta\omega, \qquad \mathbf{t} = \omega/|\omega|, \tag{7.1}$$

for any curve with $D\omega/Dt = S\omega + \nu\Delta\omega$; `inner_perpPart` shows the orthogonal strain is invisible to the stretching rate $\mathbf{t}\cdot S\mathbf{t}$. The hidden-property hypothesis `PeriodicBendingRateHiddenProperty` (coined; new formulation dated Oct 31 2026; precursors (geometric only, no regularity claims): the author's May 17/21 2026 notes ($\Pi_u$ orthogonal to $u$, "invisible to the dynamics generated by $H$") and Aug 10 2026 notes *The Frohmanian Symplectic Tether* ($B_{\text{anti}}$ via $\hat{u}\times(\cdot)$, $B_{\text{sym}}$ with pointwise $\Pi_u$)) is the conjunction of `PeriodicOrthogonalStrainControl` and `OrthogonalStrainToDirectionHolder`, both **open**; `periodicDirectionRegularity_of_bending` (proved) derives `PeriodicDirectionRegularity` from it. Classical step broken down: `abs_cfDepletion_le` (proved) bounds the Constantin–Fefferman kernel factor $(e\cdot a)\det(e,a,b)$ by $|b-a|$, and `cfDepletion_le_of_holder` by $K|x-y|^{1/2}$; `gronwall_integrating_factor` (proved) is Grönwall with rate $g = G' \in L^1$; `periodicDirectionCriterion_of_growthL1` (proved) gives `PeriodicDirectionCriterion` from the weaker open `PeriodicDirectionEnstrophyGrowthL1` (Biot–Savart stretching representation on $\mathbb{T}^3$ + BdV–B interpolation, not formalized) and `PeriodicH1BlowupCriterion`.

- Leray and analytic CF breakdown (`ClassicalSteps.lean`): `hsSq_bounded_of_rate` (proved; linear rate bound ⇒ bounded $\|u\|_{H^s}^2$ on $[0,T^*)$); `periodicH1BlowupCriterion_of_smoothing` (proved) derives `PeriodicH1BlowupCriterion` from the open parabolic-smoothing rates `PeriodicH2SmoothingRate` ($H^1 \to H^2$) and `PeriodicH3SmoothingRate` ($H^2 \to H^3$) together with the proved $H^3$ blow-up alternative; `periodicDirectionEnstrophyGrowthL1_of_holderStretching` (proved) derives `PeriodicDirectionEnstrophyGrowthL1` from the open `PeriodicHolderStretchingEstimate` (Biot–Savart stretching on $\mathbb{T}^3$ + BdV–B) and `PeriodicEnstrophyTimeIntegrable` (energy equality); `periodicBKMIntegrability_of_classical_bending` (conditional).

- Energy equality and the standard smoothing form (`EnergyEquality.lean`, Oct 31 2026; all names coined). `energy_hasDerivAt` (proved) is the energy equality

$$\frac{d}{dt}\|u\|_{L^2}^2 = -8\pi^2\nu\big(\|u\|_{H^1}^2 - \|u\|_{L^2}^2\big) \quad \text{on } (0,T^*), \tag{7.2}$$

from the unconditional $H^s$ identity at $s = 0$ and `PeriodicNonlinearEnergyCancellation`, the Fourier form of $\langle(u\cdot\nabla)u,u\rangle = 0$ [9, 13]. This cancellation is now *proved* unconditionally (`EnergyCancellation.lean`, `periodicNonlinearEnergyCancellation`): on the Fourier side the Leray projector drops out of the pairing with a divergence-free field, the coefficient decay from $u \in H^8$ makes the double sum absolutely summable, and the involution $(m,j)\mapsto(m,-(m+j))$, together with Hermitian symmetry and $k\cdot\hat{u}(k) = 0$, negates each term, so the real part vanishes. Consequently `periodicEnstrophyTimeIntegrable` proves `PeriodicEnstrophyTimeIntegrable` with no hypothesis. `periodicEnstrophyTimeIntegrable_of_cancellation` (proved) then discharges `PeriodicEnstrophyTimeIntegrable`: with $E(t) = \|u(t)\|_{L^2}^2$


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antitone on $[0, T^*)$ and extended to $T^*$ by its left limit, $G(t) = \int_0^t E - E(t)/(8\pi^2\nu)$ is continuous on $[0, T^*]$ with $G' = \|u\|_{H^1}^2$. *Discrepancy with the standard estimate.* The hypothesis `PeriodicH2SmoothingRate` requires $\frac{d}{dt}\|u\|_{H^2}^2 \le K(1 + \|u\|_{H^1}^2)^m \|u\|_{H^2}^2$ pointwise. The standard 3D estimate [14, 11] gives, after absorbing the viscous $H^3$ term, a coefficient with one extra factor $\|u\|_{H^2}$:

$$\frac{d}{dt}\|u\|_{H^2}^2 \le \frac{C}{\nu}\|u\|_{H^1}\|u\|_{H^2} \cdot \|u\|_{H^2}^2, \tag{7.3}$$

and closes only because $\int_0^T \|u\|_{H^2}^2\, dt < \infty$ when $\|u\|_{H^1}$ is bounded. The pointwise form is not known to hold. It is retained in the development, marked accordingly, and superseded by `PeriodicH2SmoothingRateStd` (the display above) and `PeriodicH2RootTimeIntegrable` ($t \mapsto \|u\|_{H^2}$ has a continuous primitive on $[0, T^*]$ when $\|u\|_{H^1}$ is bounded), both open; `periodicH1BlowupCriterion_of_smoothingStd` (proved) gives `PeriodicH1BlowupCriterion` from these two and `PeriodicH3SmoothingRate`. The $H^3$ Prop matches the standard estimate ($m = 1$, via $|\langle [J^3, u\cdot\nabla]u, J^3 u\rangle| \lesssim \|u\|_{H^2}\|u\|_{H^3}\|u\|_{H^4}$). `periodicBKMIntegrability_of_classical_bending_std` (conditional) assembles the chain.

- $\kappa = 0$ tether (`TetherKappaZero.lean`): `tetherForm_zero`, `tetherGeometricControl_zero`, `tetherImpliesDirection_zero_iff` (proved): at $\kappa = 0$ the tether control is vacuous, so "tether control ⇒ direction regularity" is *equivalent* to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_tetherKappaZero` (conditional).

## 8 Status

- **Proved, unconditional:** local existence ($\nu \ge 0$), uniqueness, coefficient ODE and decay, $H^s$ energy identity, Parseval, physical form of the $H^3$ rate, Kato–Ponce toolkit including `lower_term_bound` and `top_term_zero`, the $s = 3$ commutator bound `periodicPhysicalCommutatorBound`, the blow-up alternative, the low-frequency bound `periodicLowFreqVorticityBound`, $H^3$ BKM control `periodicBKMHsControl_three` and the BKM criterion `periodicBKMCriterion`, the equivalence `periodicBKMIntegrability_iff_globalLifespan`, and the $H^1$ enstrophy identity `periodicEnstrophyIdentity`.
- **Conditional:** Statement (B) is proven conditionally on `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). Small zero-mean $H^1$ data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_bending` gives integrability from `PeriodicDirectionEnstrophyGrowthL1`, `PeriodicH1BlowupCriterion` and the open `PeriodicBendingRateHiddenProperty`.
- **Direction route, $L^1$ form:** `periodicDirectionCriterion_of_growthL1` gives `PeriodicDirectionCriterion` from `PeriodicDirectionEnstrophyGrowthL1` and `PeriodicH1BlowupCriterion`; these in turn reduce (proved) to `PeriodicHolderStretchingEstimate`, `PeriodicH2SmoothingRateStd`, `PeriodicH2RootTimeIntegrable` and `PeriodicH3SmoothingRate`, all open here (`PeriodicNonlinearEnergyCancellation` and `PeriodicEnstrophyTimeIntegrable` are proved) (the older `PeriodicH2SmoothingRate` is stronger than the standard estimate; see above). The bending-rate hypothesis `PeriodicBendingRateHiddenProperty` (coined Oct 31 2026, open) implies `PeriodicDirectionRegularity`; the identity $D\mathbf{t}/Dt = (I - \mathbf{t}\otimes\mathbf{t})S\mathbf{t} + \nu|\omega|^{-1}(I - \mathbf{t}\otimes\mathbf{t})\Delta\omega$ is proved.


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- **Side routes:** every Prop listed under the direction, bending, small-data, smoothing and tether routes above is open (or classical but not formalized) and is *not* an argument of `clay_B_exact_of_integrability`.
- **Open:** `PeriodicBKMIntegrability` — the sole open input to `clay_B_exact_of_integrability`. Since `periodicBKMIntegrability_iff_globalLifespan` proves it equivalent to `PeriodicGlobalLifespan`, it carries the full content of the Millennium problem; it is not a smaller residual step. Neither the Clay problem nor global regularity of 3D Navier–Stokes is claimed.

## Reproducibility

Run `lake build` and then `lake env lean AxiomAudit.lean`. The build contains no `sorry`, and the audit reports only `propext`, `Classical.choice` and `Quot.sound`.

**Counting convention.** At commit `36fa434`, `AxiomAudit.lean` reports 1048 non-internal theorem constants and 285 definitions in the `PeriodicNSContinuation` module tree (kernel count, includes automatically generated theorems). Figures 755 and 867 in earlier versions referred to the source and kernel counts at commit `f9a26c7` and are superseded.

**Source.** B. S. Frohman, *periodic-ns-classical-continuation*, Lean 4 repository, `https://github.com/BenFrohman/periodic-ns-classical-continuation`, commit `36fa434` (`https://github.com/BenFrohman/periodic-ns-classical-continuation/tree/36fa434`).

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## A Axiom audit

Commit `36fa434`; command `lake build && lake env lean AxiomAudit.lean`. The module tree `PeriodicNSContinuation` contains 1048 theorems and 285 definitions; every declaration depends only on `propext`, `Classical.choice` and `Quot.sound`. No `sorryAx`, no `axiom` declaration. Open content enters only as explicit `Prop` hypotheses. All names below are in namespace `PeriodicNSContinuation`; `#print axioms` gives exactly the three standard axioms for each.


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| Declaration | Content | Hypothesis Props |
|---|---|---|
| `periodicLocalExistence_of_nonneg` | local existence, $\nu \geq 0$ | none |
| `Corrected.periodicEnergyDifferenceEstimate` | energy-difference estimate | none |
| `Corrected.periodicVelocityUniqueness_of_energy` | uniqueness | none |
| `Corrected.periodicMaximalSolution_of_uniqueness` | maximal solutions | none |
| `periodicHsBlowupAlternative` | $H^s$ blow-up alternative, $s > 5/2$ | none |
| `periodicPhysicalCommutatorBound` | $s = 3$ commutator bound | none |
| `periodicLowFreqVorticityBound` | low-frequency vorticity bound | none |
| `periodicBKMHsControl_three` | $H^3$ BKM control | none |
| `periodicBKMCriterion` | BKM criterion | none |
| `periodicBKMIntegrability_iff_globalLifespan` | `PeriodicBKMIntegrability` $\Leftrightarrow$ `PeriodicGlobalLifespan` | none (both sides open) |
| `clay_B_exact_of_globalLifespan` | `PeriodicGlobalLifespan` $\Rightarrow$ `Clay_B_exact` | `PeriodicGlobal Lifespan` |
| `clay_B_exact_of_integrability` | Statement (B), conditional | `PeriodicBKM Integrability` only |

Conditional theorems carry their hypotheses as arguments; the kernel audit is unaffected by them. This paper does not claim a solution of the Clay problem or global regularity of 3D Navier–Stokes.


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